
Factory Analytics Web App Architecture

Comprehensive technical overview of React-based systems, data flow management, and production lifecycle integration.

ENTRY POINT & SYSTEM PROVIDERS



src/index.js

The application bootstrap point using **React 18** with `createRoot`, ensuring modern concurrent rendering capabilities.



QueryClientProvider

Managed by **React Query**, this provides a global cache for server-side state, API synchronization, and automatic refetching.



ContextProvider

Custom context managing **UI/Theme state**: Sidebars, Dark/Light mode preferences, and current user UI context.

SECURE AUTHENTICATION FLOW

-  **Cookie-Based Auth:** Tokens are stored in HTTP Cookies for persistent sessions rather than less secure localStorage.
-  **Admin-Only Access:** System validates `userType === 'admin'` before allowing entry to dashboard and protected routes.
-  **Auth Persistence:** On initial load, `App.js` reads the token; if missing, it forces redirection to the login interface.
-  **Global Logout:** Token clearance collapses the global sidebar/navbar, resetting the application to its minimal unauthenticated state.

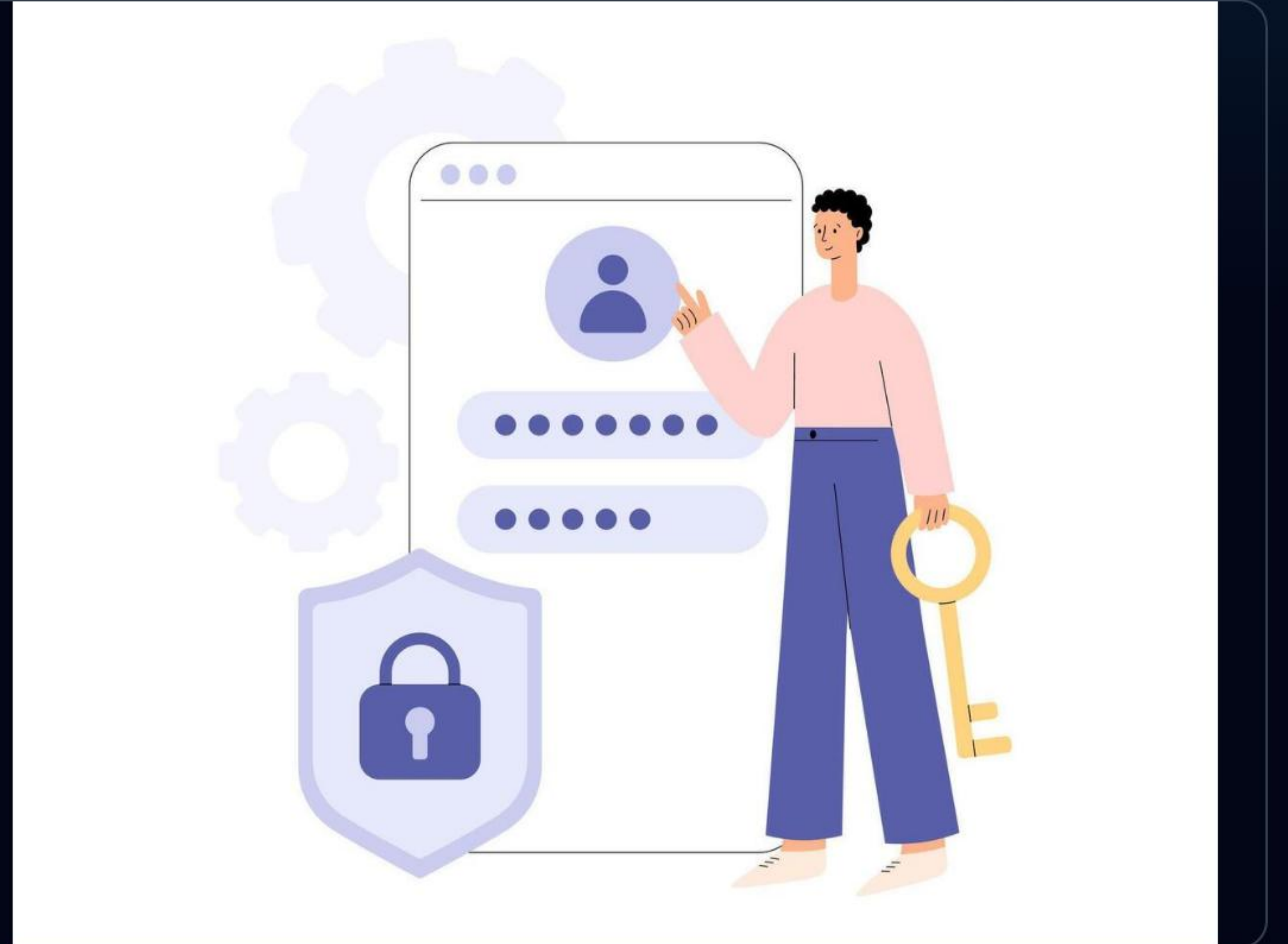
ADMINISTRATIVE LOGIN MECHANISM

The useLogin() Mutation

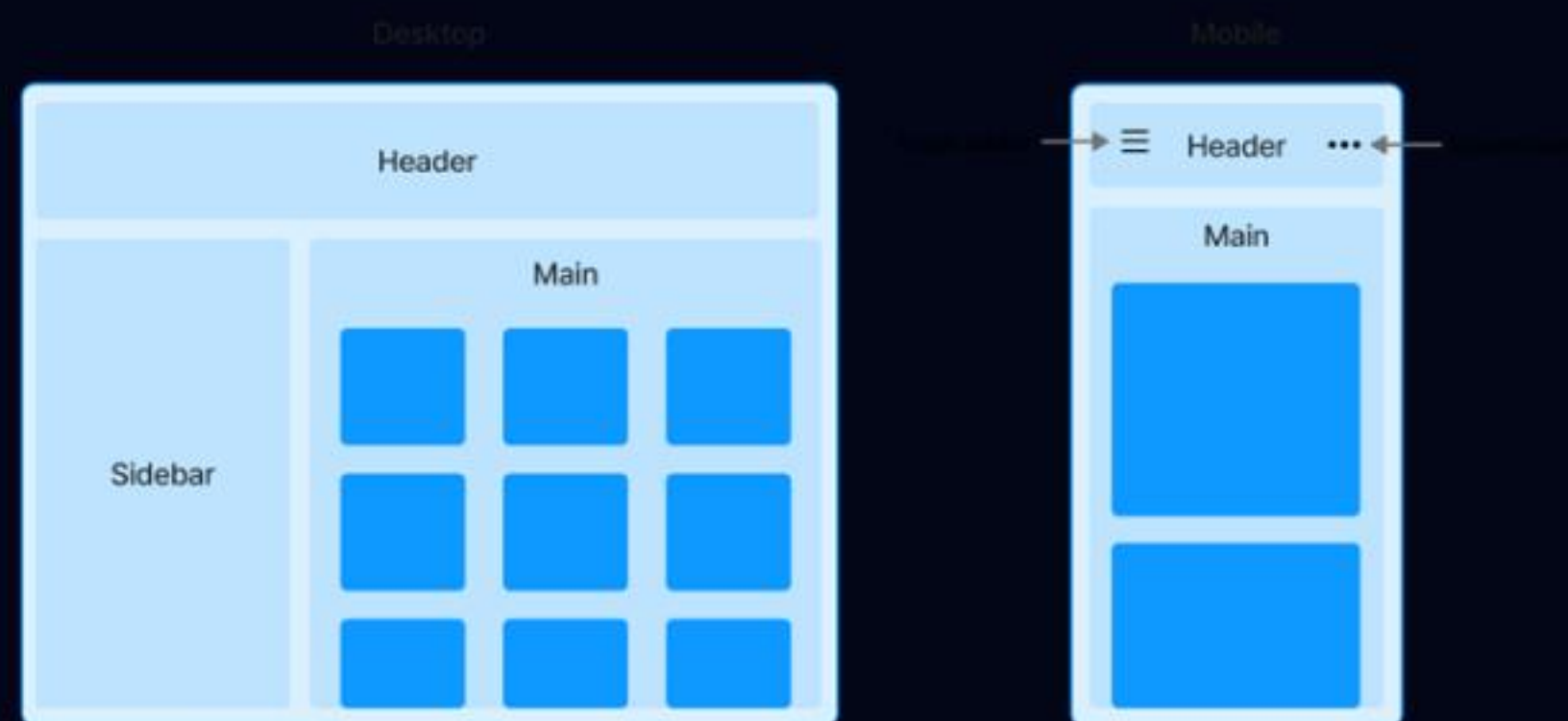
Leveraging React Query mutations, the login flow performs a secure POST request to the centralized backend endpoint.

- ✓ 30-Day token expiry set on success
- ✓ LocalStorage stores user_Id
- ✓ Automatic redirect to /dashboard

Non-admin users are greeted with specific **"Invalid User"** toast notifications to maintain system integrity.



APPLICATION ROUTING & GLOBAL LAYOUT



React Router v6

Uses BrowserRouter to handle protected and public routes, ensuring deep-linking works across factory reports.

Adaptive Layout

Sidebar visibility and width dynamically adjust based on `activeMenu` state and screen resolution via the global Context.

DUAL-STATE MANAGEMENT STRATEGY



Server State (React Query)

Handles data from **Axios** calls. Manages caching, background fetching, and synchronization for orders, materials, and schedules.



Client State (React Context)

Manages ephemeral UI data like **ThemeMode** (Light/Dark), screenSize, and sidebar interaction states.

API CONNECTIVITY & LAYERING

451

EXPIRY STATUS HANDLER

100%

CENTRALIZED ENDPOINTS

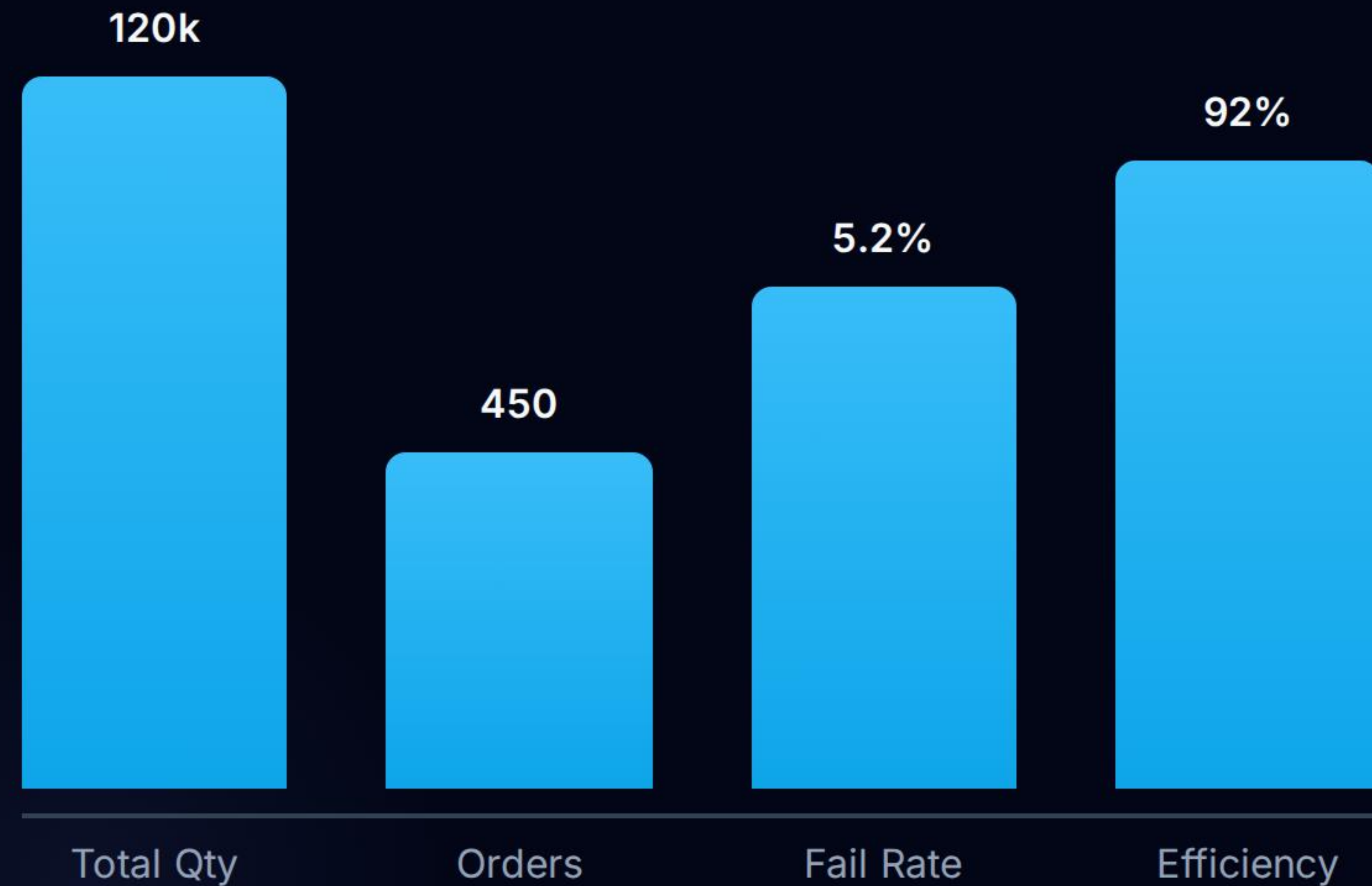
The Axios Instance Layer

All calls utilize a single axiosInstance configured with a baseURL from environment variables.

Interceptors:

- **Request:** Automatically injects Bearer token from cookies.
- **Response:** Monitors for 401/451 errors to trigger automatic re-authentication flows.

REAL-TIME DASHBOARD ANALYTICS



Data Pipeline

The dashboard fetches live KPIs from `/api/admin/orders/total-orders-quantities`.

It utilizes a **TabbedView** architecture to separate current orders from historical performance graphs.

ORDER CLASSIFICATION & MANAGEMENT



Customer Orders

Filtered view excluding 'self production' entries. Tracks external client demands and delivery timelines.



Self Production

Dedicated internal production tracking. Focuses on stocking and inventory-led manufacturing orders.



CRUD Workflow

Integrated Formik + Yup forms for adding and editing orders with real-time API state updates.

MANUFACTURING PRODUCTION LIFECYCLE



INTEGRATED UI COMPONENT LIBRARY



Syncfusion

Enterprise-grade Grids, Calendars, and Dropdowns for high-density data management.



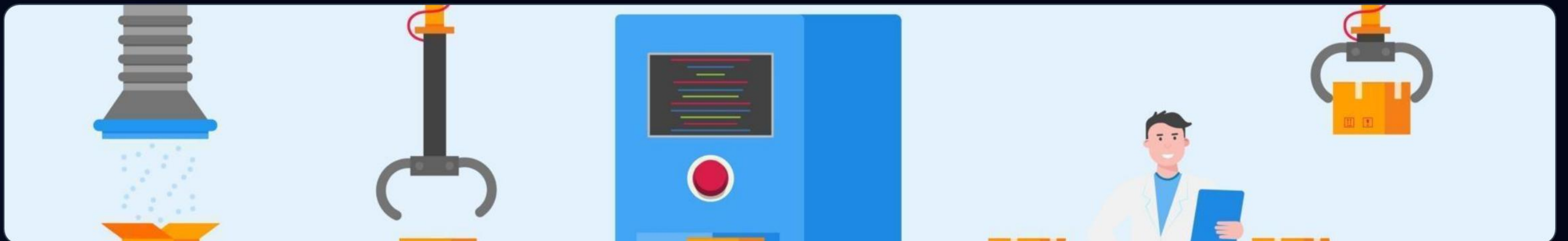
Formik & Yup

Robust form handling with schema-based validation for all administrative inputs.



React-Toastify

Asynchronous feedback system for API successes and permission errors.





Questions & Answers

Technical Architecture Review | Build & Deployment Strategy

Dev: npm start | **Prod:** npm run build

Environment: Managed via REACT_APP_BASE_URL

IMAGE SOURCES



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